Yuancheng [Mike] Luo

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PROFILE

Research scientist with a strong background in digital signal processing, optimization, and audio-acoustics. Held previous roles in spatial audio research and development, audio effects plugin software development, and non-parametric machine learning models for audio personalization research.

EDUCATION

2009 – 2014 University of Maryland College Park, College Park, MD
Ph.D. and M.Sc. in Computer Science

2006 – 2009 University of Maryland College Park, College Park, MD
B.Sc. Double major in Computer Science and Mathematics

WORK EXPERIENCE

2018 – 2026 Amazon Lab126, Cambridge, MA
Research Scientist. Published, patented, and productized novel algorithms for optimal acoustic beamformer designs, flexible loudspeaker arraying for home-theater, constraint optimized audio limiting, cross-talk cancellation, up-mixing, filter-fitting, head-tracked binaural virtualization, and efficient loudspeaker-microphone-room acoustic simulation. Developed Matlab tools for beamformer and filter design and realization. Architected and tuned the spatial audio playback software deployed on Amazon's Echo product line, and Alexa home theater.

2017 – 2018 **NuSpace Audio**
Solo Founder/Research Scientist/Developer. Founded spatial audio software startup for music and sound post-production digital audio workstation plugin. Designed novel algorithms for simulating reverberation and direction cues in high-dimensional spaces. Developed audio plugins for binaural 3D audio panning and reverberation, binaural multi-effects, and Ambisonics reverberator plugins. Launched, designed, marketed, and demoed audio products and website.

2014 – 2016 Visisonics Corporation, College Park, MD
Research Scientist. Developed algorithms for personalizing spatial audio. Developed generative models for synthesizing and imputing head-related transfer functions (HRTFs). Designed graphical interfaces (JUCE framework), signal process routines, and metrics for the real-time editing and rendering of HRTFs. Integrated RealSpace3D libraries with game engines (Audiokinetic Wwise, Unity, Unreal 4).

2009 – 2014 University of Maryland Institute for Advanced Computer Studies, College Park, MD
Graduate Research Assistant. Developed machine learning (Bayesian Regression, Gaussian and random processes, non-negative matrix factorizations, neural networks) algorithms for spatial audio reproduction (binaural rendering, DSP, psychoacoustics). Designed HRTF recommendation system via active-learning algorithms with listening tests, Gaussian process models for fast HRTF spatial interpolation, sparse non-negative algorithms for fast HRTF convolution. Parallelized non-negative least squares (NNLS) problems on the GPU.

2007 – Summer University of Maryland Institute for Advanced Computer Studies, College Park, MD
Undergraduate Researcher (Internship). Researched and parallelized Canny edge detection and kernel density estimation algorithms on the GPU for computer vision front-end and background subtraction. Developed middleware to interface between C/CUDA and Fortran languages.

SOFTWARE SKILLS, OPEN-SOURCE, AND PATENTS

Languages/APIs: Matlab, Simulink, C/C++, Microsoft Visual Studios,, CUDA, OpenGL, OpenAL, VST/AU

Open-source: Liquidus GPU based fluid dynamics (Navier-Stokes) sandbox simulation game

Patents: US11128953B2, US11830471B1, US10237676B2, US9681250B2

SELECTED PUBLICATIONS

1. Yuancheng Luo, D. Yamkovoy and G. Garcia, "Constraint Optimized Multichannel Mixer-Limiter Design," ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2026, pp. 15692-15696.

2. Yuancheng Luo, "Optimized Loudspeaker Panning for Adaptive Sound-Field Correction and Non-stationary Listening Areas," 2025 Journal of the Audio Engineering Society, AES Convention 159, 385.

3. Yuancheng Luo, "Constant Directivity Loudspeaker Beamforming," 2024 32nd European Signal Processing Conference (EUSIPCO), Lyon, France, 2024, pp. 246-250.

4. Yuancheng Luo, "Active Barycentric Beamformed Stereo Upmixing," 2023 31st European Signal Processing Conference (EUSIPCO), Helsinki, Finland, 2023, pp. 316-320.

5. Yuancheng Luo, "Spherical harmonic covariance and magnitude function encodings for beamformer design," Journal on audio speech and music processing, 2021, 41.

6. Yuancheng Luo, Wontak Kim, "Fast Source-Room-Receiver Acoustics Modeling," 2020 28th European Signal Processing Conference (EUSIPCO), Amsterdam, Netherlands, 2021, pp. 51-55.

7. Yuancheng Luo, Dmitry N. Zotkin, and Ramani Duraiswami. "Gaussian Process Models for HRTF based 3D Sound Localization", International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Florence, Italy, 2014, pp. 2858-2862.

8. Yuancheng Luo, Dmitry N. Zotkin, and Ramani Duraiswami. "Virtual Autoencoder based Recommendation System for Individualizing Head-related Transfer Functions", IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), 2013, New Paltz, NY.

9. Yuancheng Luo, Dmitry N. Zotkin, and Ramani Duraiswami, "Gaussian Process Data Fusion for Heterogeneous HRTF Datasets", IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), 2013, New Paltz, NY.

10. Yuancheng Luo, Dmitry N. Zotkin, Hal Daumé III and Ramani Duraiswami, "Kernel Regression for Head-Related Transfer Function Interpolation and Spectral Extrema Extraction", Proceedings 38th International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Vancouver, 2013.

11. Yuancheng Luo, Ramani Duraiswami, "Fast Near-GRID Gaussian Process Regression", AISTATS 2013.

12. Yuancheng Luo, Dmitry N. Zotkin, and Ramani Duraiswami, "Statistical Analysis of Head-Related Transfer Function (HRTF) data", International Congress on Acoustics, Montreal, Proceedings of Meetings on Acoustics, 2013.

13. Yuancheng Luo, Ramani Duraiswami, "Efficient Parallel Non-Negative Least Squares on Multicore Architectures", SIAM Journal on Scientific Computing 2011.

14. Yuancheng Luo, Ramani Duraiswami, "Canny Edge Detection on NVIDIA CUDA," Proceedings of the Workshop on Computer Vision on GPUS, CVPR 2008.